



Silicon segregation enhances IASCC of austenitic stainless steels via oxide degradation: Decoupling from chromium depletion effects

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ABSTRACT

Grain boundary (GB) segregation plays a critical role in irradiation-assisted stress corrosion cracking (IASCC) by influencing both intergranular oxidation and GB cohesive strength. To decouple the effects of Si enrichment and Cr depletion on crack enhancement, four model alloys with systematically varied Si and Cr concentrations were designed, and slow strain rate testing (SSRT) was performed on these alloys. The results demonstrate that the cracking susceptibility increases significantly with increasing Si content. In contrast, the low-Cr alloy exhibited only a marginal increase in cracking susceptibility compared to the alloy containing 0.85 wt% Si. These findings indicate that Si segregation at GBs is a more important factor promoting IASCC.

1. Introduction

Irradiation-assisted stress corrosion cracking (IASCC) in materials, which occurs early in the life of PWR core components (1–3 dpa), has become a major degradation mechanism of light water reactors (LWRs) [1,2]. IASCC is predominantly intergranular, making it somewhat more difficult to detect via non-destructive testing (NDT) methods and often leads to unexpected consequences. To date, failures have been reported in over 20 core components-manufactured from stainless steels and nickel-base alloys, including baffles, former bolts, fuel rods, control blades, instrumentation tubes, and springs-in both pressurized water reactors (PWRs) and boiling water reactors (BWRs) [3,4]. These incidents have significantly impaired economic efficiency and posed a substantial safety risks to nuclear reactor power plants.

Neutron irradiation is responsible for the premature or enhanced cracking of reactor core components, as it concurrently induces microstructural and microchemical changes in materials, along with water radiolysis. Microstructure alterations, such as the formation of black dots, dislocation loops, voids, and precipitates, lead to hardening and increased strength, which have been correlated with IASCC susceptibility [5–7]. Microchemical changes typically involve elemental segregation, occurring both at grain boundaries (GBs) and within grains (e.g., in clusters and precipitates). In austenitic stainless steels, segregation of

both minor and major elements has been observed at GBs, including enrichment of Ni, Mn, Si, and P, and depletion of Fe, Cr, and Mo [8–11]. Extensive research on the microstructural and microchemical evolution of irradiated materials, as well as their effects on stress corrosion cracking (SCC) behavior, has yielded significant progress in understanding IASCC mechanisms. Grain boundary oxidation is widely recognized as a precursor to IASCC crack initiation [12,13], as it weakens GBs and reduces fracture resistance [14–17]. Applied stress accelerates the oxidation process and promotes localized deformation in grains with favorable orientations, further enhancing oxidation at the intersection of dislocation channels (DCs) and GBs [14–17]. When DCs terminate at GBs, they generate high local stress concentrations, sufficient to fracture the oxidized GBs, thereby initiating cracks [3,18–20].

Given the critical roles of GB oxidation and fracture in IASCC, considerable attention has been directed toward understanding elemental segregation at GBs, particularly Si enrichment and Cr depletion. Studies on sensitized stainless steels have demonstrated that GB Cr depletion can lead to severe SCC in oxygenated environment [21,22] as insufficient Cr prevents the formation of a protective Cr-rich oxide film. In contrast, the GB carbides formed during sensitization can inhibit SCC cracking in hydrogenated high-temperature water [23]. Given that the Cr depletion feature at GB induced by irradiation differs from that of thermal sensitization (which involves carbide precipitation), its effect

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Table 1

The chemical composition of four model alloys (wt%).

Alloy	Fe	Cr	Ni	Mn	Si	C	S
H1	Bal.	19.84	24.94	2.25	0.85	< 0.03	< 0.03
H2	Bal.	19.90	25.06	2.26	2.63	< 0.03	< 0.03
H3	Bal.	18.29	23.06	2.10	4.08	< 0.03	< 0.03
H4	Bal.	8.32	25.83	2.22	0.33	< 0.03	< 0.03

on SCC cracking in a simulated PWR environment is expected to be distinct. Post-irradiation annealing which desegregates Cr and Si from GBs has been correlated with reduced IASCC susceptibility [24–26]. SCC tests on high-Si model alloys demonstrated that both crack initiation [27,28] and crack propagation rates [29–32] increase with Si concentration. Although variations in Si (~1–5 %) and Cr (12–19 %) content have been shown to exert limited influence on the electrochemical behavior of materials in high-temperature water [33–35], irradiated materials exhibit deeper GB oxidation compared to their unirradiated counterparts [13,36], a phenomenon critical to IASCC.

However, it remains unclear whether the preferential GB oxidation is driven primarily by Si or Cr, as their segregation occurs simultaneously. Previous studies employed model alloys with either low Cr/high Si or variable Si/normal Cr, failing to effectively decouple the two effects. Therefore, systematic experiments on two sets of model alloys are necessary: one with a Cr content comparable to that of 304 stainless steel and varying Si levels, and another with low Cr and normal Si. Such studies would fill the knowledge gap and contribute to a more comprehensive understanding of IASCC mechanisms.

2. Experimental

2.1. Materials and specimen preparation

To isolate the effects of Cr depletion and Si enrichment on IASCC, four model alloys with systematically varied Si and Cr content were designed, based on the GB chemical composition of commercial nuclear-grade 304 stainless steel after neutron irradiation. The Ni concentration was fixed at 25 wt% in all alloys to simulate the Ni enrichment typically observed under high dpa damage conditions. For alloys with varying Si concentration (1, 2.5, 4.0 wt%), the Cr concentration was maintained at 19 wt%. The fourth alloy, with reduced Cr content (9 wt%), was

designed to have a Si level of 0.33 % comparable to that of standard nuclear-grade 304 stainless steel. The Mn concentration was consistently maintained at 2.2 wt% across all alloys.

Columnar ingots were prepared using a vacuum induction levitation melting furnace. These ingots were then forged at 1150 °C into cuboids with a thickness of approximately 50 mm, followed by quenching in ice water. To verify the target compositions, chemical analyses were conducted using a wavelength-dispersive X-ray fluorescence spectrometer, with results summarized in Table 1. Grain size characteristics were evaluated via electron backscatter diffraction (EBSD), and data were processed using AZtecCrystal software. The resulting inverse pole figure (IPF) maps and grain size distributions are presented in Fig. 1. All alloys exhibited comparable average grain sizes, ranging from 44 to 71 μm.

Each alloy block was sectioned midway through its thickness. One half underwent an additional one-step forging at 200 °C, achieving a 20 % reduction in thickness. This deformation treatment was applied to enhance mechanical strength and increase susceptibility to SCC, thereby facilitating crack initiation within a practical experimental timeframe. Tensile specimens were fabricated from the forged plates using electrical discharge machining (EDM), with the specimen geometry illustrated in Fig. 2. These tensile specimens featured a gage length of 16 mm and a rectangular gage cross-section of $2.2 \times 1.6 \text{ mm}^2$. To ensure consistency, all specimens used for slow strain rate testing (SSRT) were extracted from equivalent positions within the deformed plates across different alloys. Both the front and back surfaces of each tensile specimen were mechanically polished up to 4000-grit SiC paper to achieve a uniform surface finish and eliminate surface-related artifacts. The side surfaces of the gauge sections were also carefully polished to remove any scratch-like defects that could act as stress concentrators.

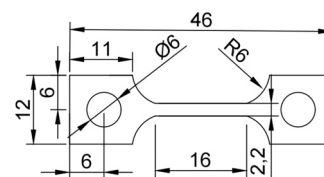


Fig. 2. Schematic diagram of the tensile specimen geometry (units: mm).

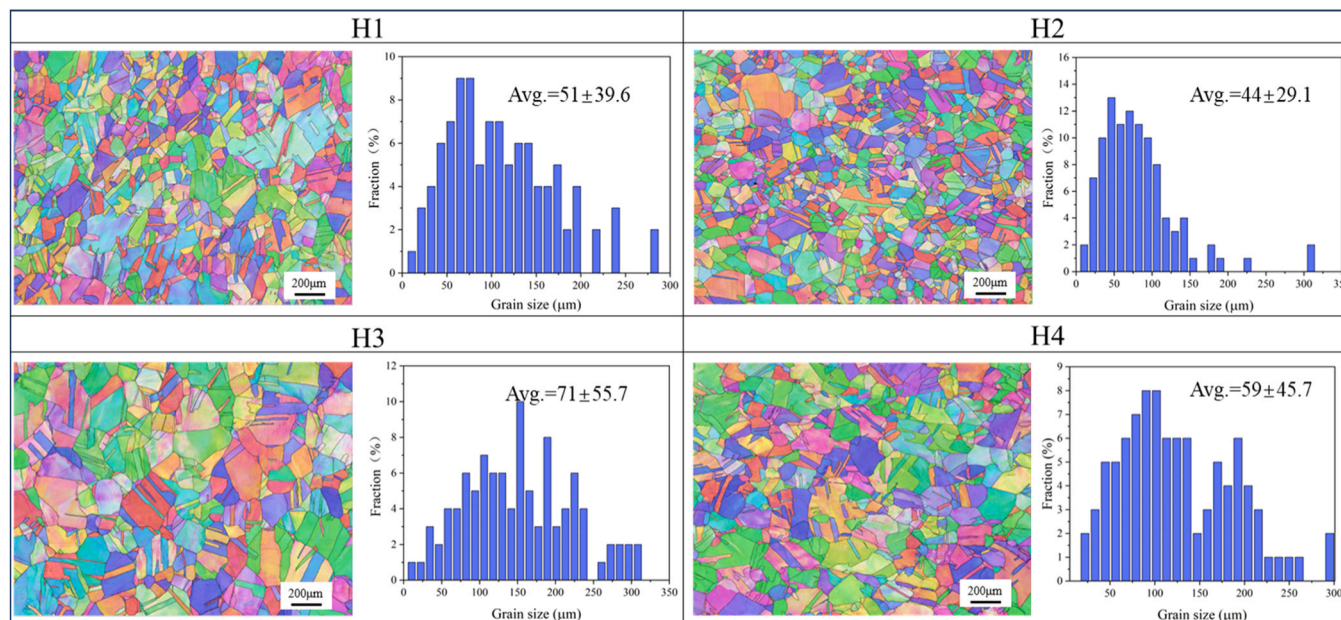


Fig. 1. Inverse pole figure (IPF) maps of the four model alloys, including grain size distributions and average grain sizes.

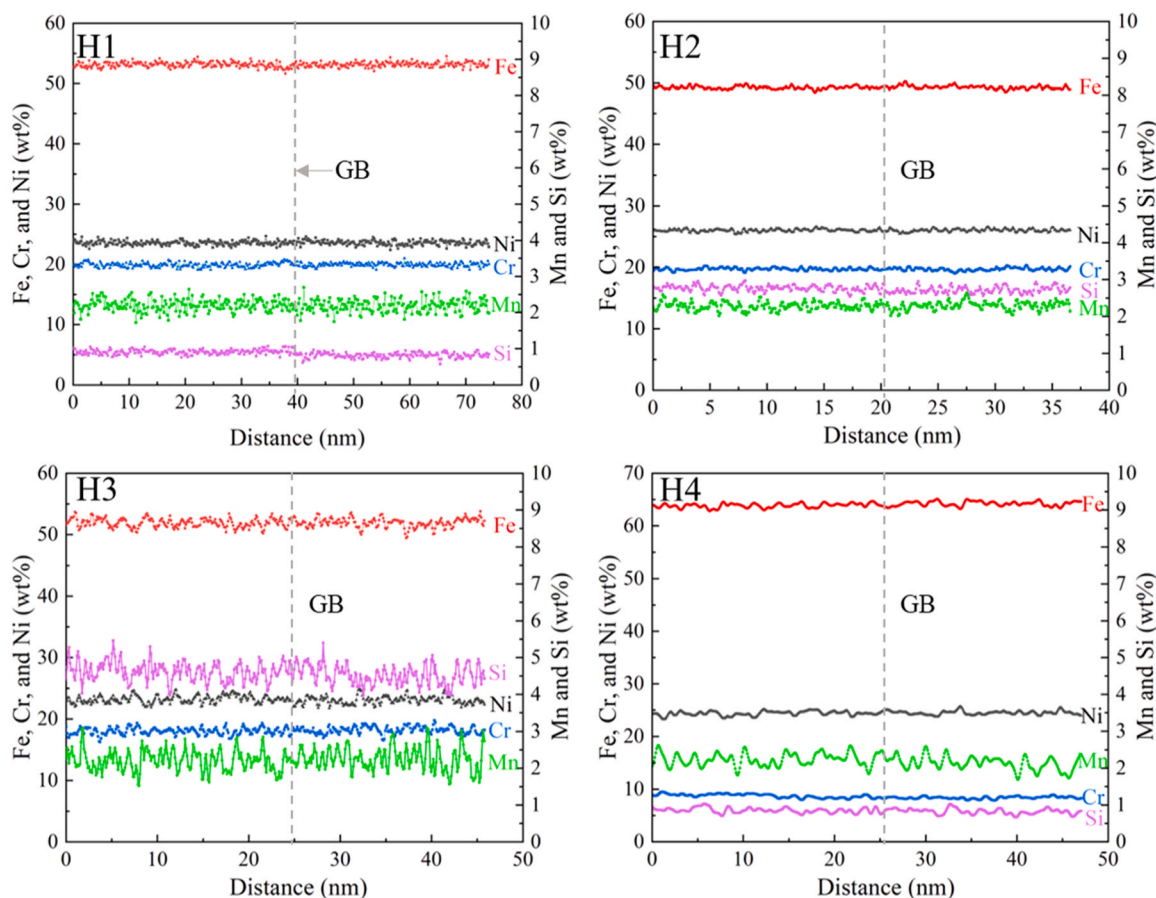


Fig. 3. Compositional profiles across a GB obtained by STEM-EDS line scan.

2.2. Grain boundary composition analysis

To investigate whether Si or Cr naturally segregates to GBs, transmission electron microscopy (TEM) specimens were prepared from each alloy. Small sections were extracted from the bulk material and mechanically polished to a thickness of approximately 100–200 μm . Discs with a diameter of 3 mm were then punched from the thinned foils and subjected to double-jet electropolishing using a Struers TenuPol-5 system. The electrolyte consisted of 5 vol% perchloric acid in methanol, maintained at a temperature below -20°C by cooling the solution with liquid nitrogen. Scanning transmission electron microscopy (STEM) coupled with energy-dispersive X-ray spectroscopy (STEM-EDS) mapping was performed using a Talos F200X G2 microscope operated at 200 kV, with a probe size of approximately 1 nm.

2.3. Slow strain rate tensile test

SSRT tests were conducted in a 316 L stainless steel autoclave connected to a freshly refreshed water loop capable of providing simulated PWR water conditions (320°C , 2.7 ppm H_2). After achieving environmental equilibrium, four specimens (one from each alloy) were strained simultaneously at a constant strain rate of $5 \times 10^{-8} \text{ s}^{-1}$. Each specimen was actuated by an independent servo motor, ensuring that the stress and strain conditions of individual samples were not affected by others. The load on each specimen and the displacement were continuously recorded via the test control program. Two fiducial marks were inscribed outside the gauge section of each specimen to enable accurate measurement of the actual plastic strain post-test. To account for system compliance and ensure consistent crack initiation across all specimens, a nominal total strain of more than 7 % was applied before stopping the tests. This strain level was determined through preliminary tests, which

confirmed it reliably induced crack initiation in all specimens under the specified environmental conditions.

2.4. Post-strain characterization

Stress–strain curves were generated from the recorded load and displacement data, from which the yield stress was derived. The actual plastic strain for each specimen was calculated based on the change in distance between the two fluidical marks before and after testing, with measurements obtained by capturing SEM images of the marks. Post-test characterization included plane-view and cross-sectional imaging of the deformed specimens using a GeminiSEM 460. Crack initiation behavior was quantified by counting the number of cracks and measuring their lengths and depths. From these measurements, crack number density, length density, and depth distribution were determined. Additionally, the morphology and composition of the oxide films were examined using SEM and EDS.

To analyze GB oxidation at crack tips, site-specific TEM specimens were prepared using a dual-beam focused ion beam (FIB)-SEM system (FEI). For deeply embedded cracks inaccessible via conventional surface FIB milling, a plane lift-out technique was employed on pre-exposed cross-sectional surfaces. The FIB lift-outs were sequentially thinned using 30 kV voltage with currents of 5 nA and 1 nA, followed by final thinning at 5 kV (60 pA), and a cleaning step at 2 kV. The resulting TEM lamellae were characterized via STEM and EDS using an FEI Talos F200X G2 microscope operating at 200 kV and equipped with a Super-X EDS detector. Quantitative EDS analysis was performed using Velox software.

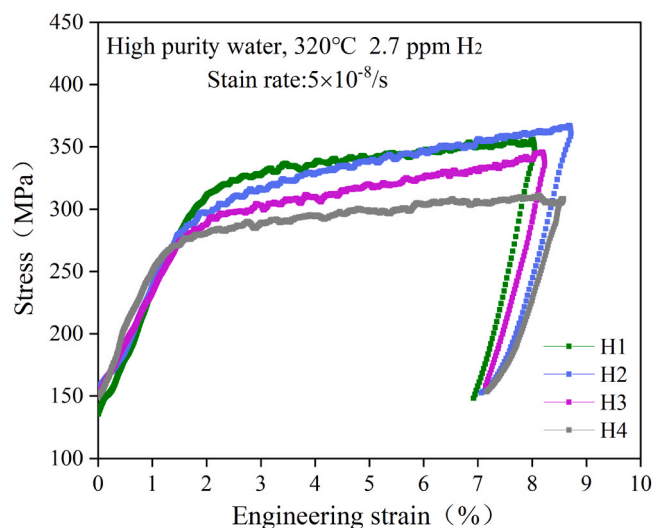


Fig. 4. Stress-strain curves of tensile samples strained at 5×10^{-8} /s in simulated PWR environment.

Table 2

Mechanical properties and strain conditions of four specimens subjected to SSRT testing in a PWR environment.

SCC parameter	H1	H2	H3	H4
Yield strength (MPa)	310	296	285	275
Plastic strain (%)	6.92	7.06	7.14	7.16
Stress at target strain (MPa)	356	367	346	315
Corrosion duration (h)	599	555	576	530

3. Results

3.1. Elemental composition of the grain boundary

Fig. 3 presents compositional profiles of the main elements across a random high-angle GB in the as-prepared model alloys. For alloys H1, H2, H3, and H4, no segregation of Si or Ni was detected at the GBs, with their concentrations remaining consistent with those in adjacent grain interiors, indicating the absence of natural segregation in these alloys. Similarly, no Cr depletion or precipitate formation was observed at the

GBs in the first three alloys. In contrast, nanoscale Cr_{23}C_6 carbides were identified along GBs in alloy H4, with corresponding Cr depletion zones evident in the regions between these precipitates. However, EDS line scans across precipitate-free GB segments in alloy H4 confirmed the absence of Cr depletion, as clearly illustrated in Fig. 3. Although nanoscale Cr_{23}C_6 carbides were observed along some GBs in alloy H4, their size was limited, and their distribution was sparse and discontinuous.

3.2. Stress-strain curves

The stress-strain curves for all tested samples are shown in Fig. 4. An initial stress of approximately 150 MPa is observed at the start of each curve, arising from the autoclave pressure acting on the tensile specimens. Key mechanical properties, including yield strength, stress at the target strain (7 %), measured actual plastic strain, and corrosion duration, are summarized in Table 2. The actual plastic strains measured after SSRT were 6.92 %, 7.06 %, 7.14 %, and 7.16 % for the four alloys, respectively, all within 5 % of the target strain value. Yield strength decreased progressively with increasing alloy number (from H1 to H4), with a maximum difference of only 35 MPa between alloys H1 and H4. The final engineering stresses at the target strain of 7 % were 356 MPa, 367 MPa, 346 MPa, and 315 MPa, respectively, with a maximum difference of 42 MPa between alloys H2 and H4. The relatively small differences in strengths among the alloys provide a well-controlled mechanical basis for evaluating and comparing their SCC susceptibility.

3.3. Oxidation and crack initiation behavior

3.3.1. Plan-view

Fig. 5 presents an overview of SEM micrographs of the gauge surfaces after SSRT, composed of stitched low-magnification images. The absence of visible necking confirms uniform elongation of all specimens. Surface-initiated cracks are marked with white arrows; although some finer cracks are not clearly resolved at this magnification, the overall distribution and quantity of cracks remain distinguishable. Alloy H3 exhibited the highest number of cracks, followed by alloy H2, while alloys H1 and H4 showed comparatively fewer cracks. To quantitatively assess SCC susceptibility, the total number of cracks in the entire gauge section was counted, and the length and width of each crack were measured. From these data, crack number density and crack length density (total crack length per unit area) were calculated. Additionally, the maximum crack width and length for each alloy were determined.

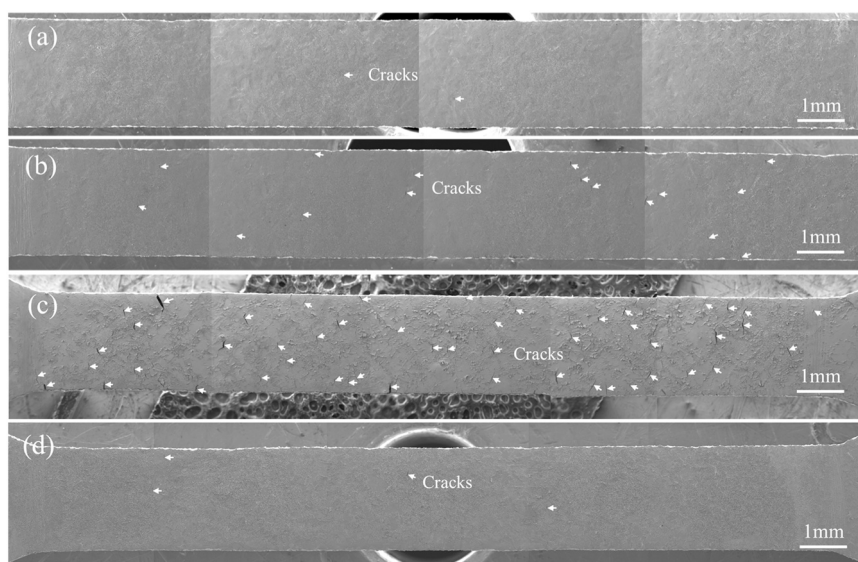


Fig. 5. Post-SSRT overview SEM micrographs of the entire gauge sections for alloys (a) H1, (b) H2, (c) H3, and (d) H4.

Table 3
Summary of SCC susceptibility metrics for four alloys after SSRT testing in a simulated PWR environment.

Alloy	H1	H2	H3	H4
Max. crack width (μm)	5.1	14.9	35.1	7.9
Max. crack length (μm)	139.6	218.8	469.5	181.6
Average crack length (μm)	71.0 ± 9.7	84.6 ± 17.6	147.4 ± 25.4	67.7 ± 6.3
Crack number density (mm ⁻²)	3.0	5.2	7.1	4.0
Crack length density (μm·mm ⁻²)	210.0 ± 29.1	439.9 ± 91.5	1046.5 ± 180.3	270.8 ± 25.2

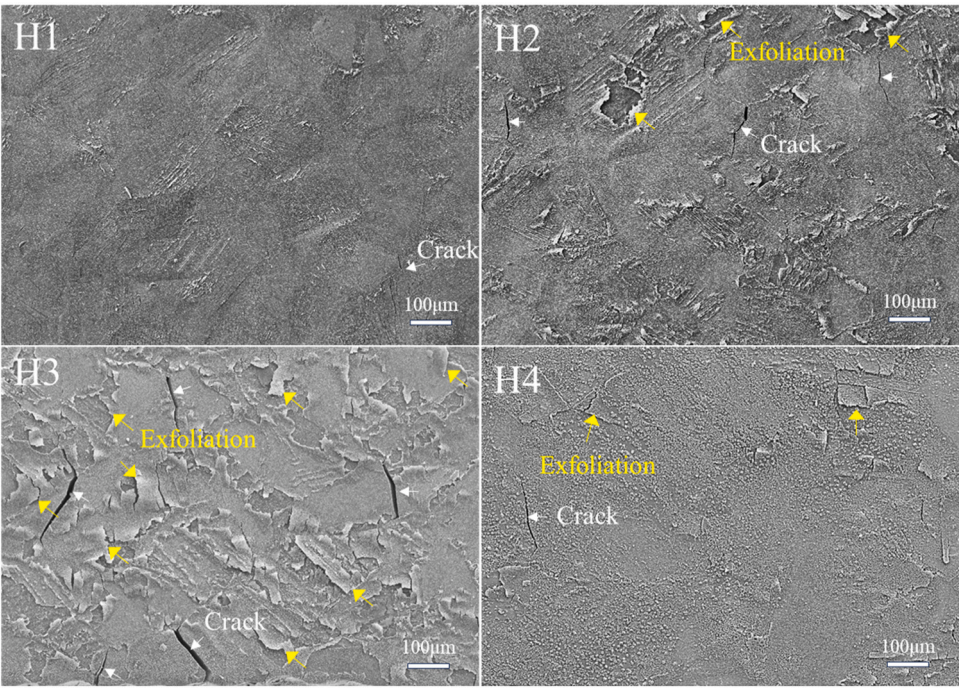


Fig. 6. SEM images showing crack initiation behavior and oxide morphology on the gauge section surfaces of alloys H1, H2, H3, and H4 after SSRT in a simulated PWR environment.

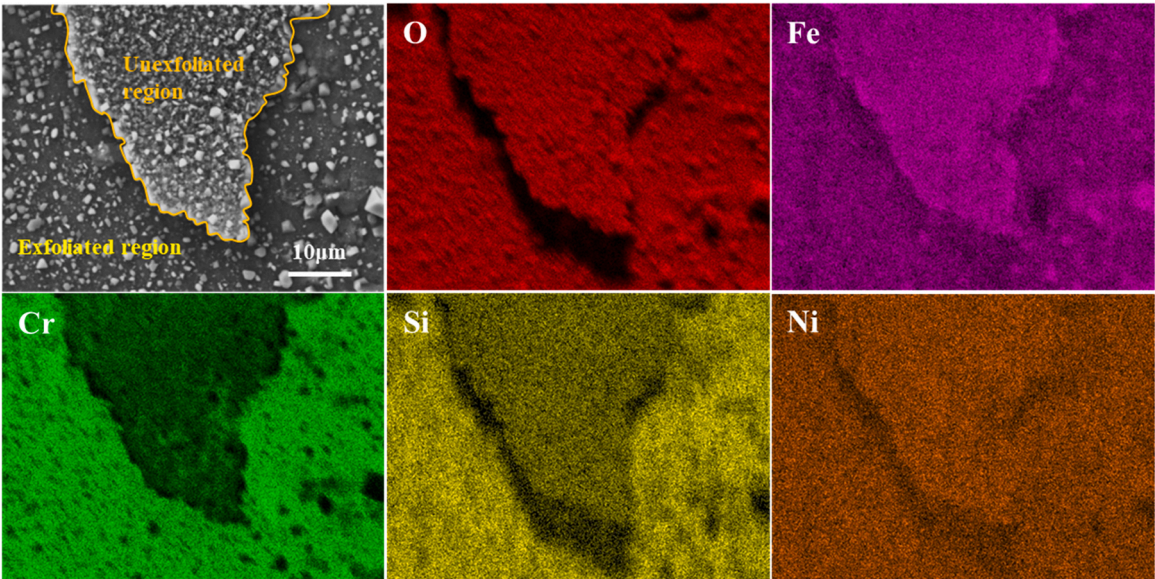


Fig. 7. SEM-EDS elemental mapping of the surface oxide film on alloy H3 after SSRT, showing secondary electron image with regions of intact and exfoliated oxide, and corresponding elemental distributions of O, Fe, Cr, Si, and Ni.

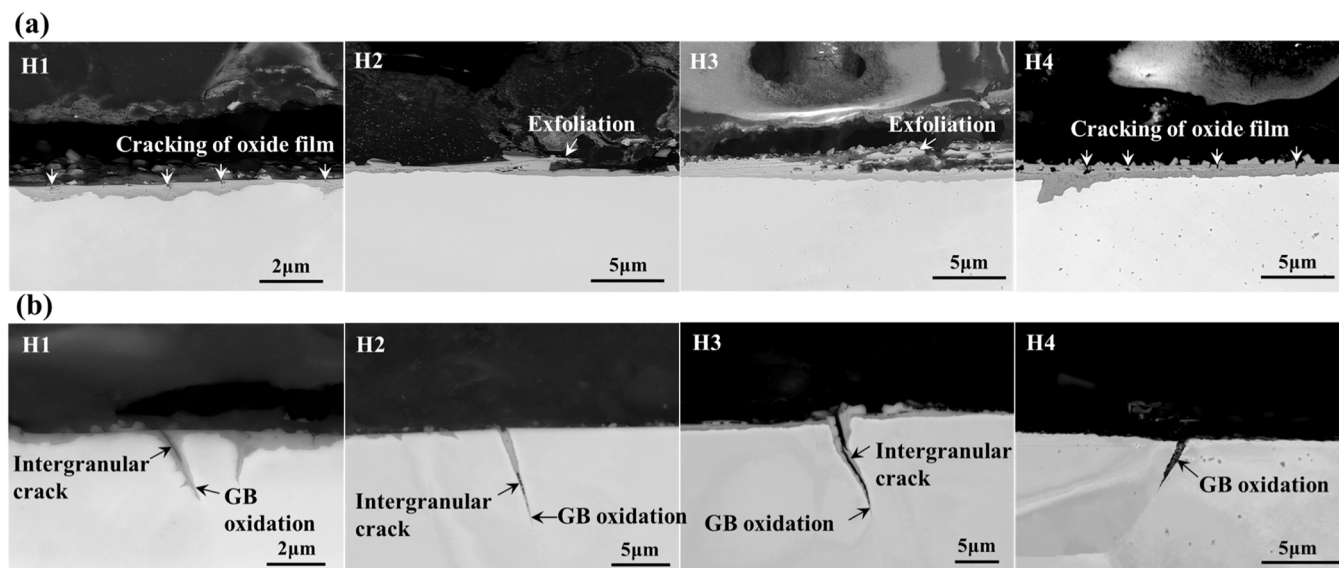


Fig. 8. Cross-sectional BSE images of the four alloys, showing (a) surface oxides and (b) oxidized GB or intergranular crack.

All quantitative metrics are summarized in Table 3. Except for average crack length, all parameters—including maximum crack width, maximum crack length, crack number density, and crack length density—increase in the order: H1 < H4 < H2 < H3. This trend indicates a corresponding increase in SCC susceptibility across the four alloys. The initiation site of stress corrosion cracking (SCC) is strongly influenced by factors such as the angle between the tensile axis and the grain trace, the Luster-Morris m' parameter, and grain boundary misorientation [37]. Consequently, in a polycrystalline sample with a sufficient number of grains, the number of initiated intergranular cracks is thus determined by the material's inherent SCC resistance.

Fig. 6 shows higher-magnification SEM images of the gauge surfaces, revealing detailed crack initiation sites and oxide morphology. Clear differences in cracking behavior and oxidation characteristics are observed among the alloys. The surface of alloy H1 remains relatively smooth and intact, with minimal oxide spallation. In contrast, alloys H2 and H3 exhibit rough, irregular surfaces with extensive peeling and flaking of oxide films, with alloy H3 showing more severe and widespread exfoliation compared to the localized spallation observed in H2. Alloy H4 displays a moderately affected surface, featuring isolated regions of high-density oxide particles and minor exfoliation. Widespread oxide exfoliation suggests poor adhesion of the oxide film or high susceptibility to fracture under tensile stress, likely due to high defect density or limited oxide ductility.

To further investigate this behavior, SEM-EDS analysis was conducted on alloy H3, which exhibited the most severe exfoliation (Fig. 7). The image includes both adherent and spalled oxide regions: the area within the orange outline represents intact oxide, which is depleted in Cr and Si; in contrast, the exfoliated region is enriched in Cr and Si, indicating a distinct compositional difference between the two oxide states. The TEM-EDS analysis on the high-Si alloy performed by Han et al. [38] have demonstrated that the inner oxide film is rich in Si and Cr, which is a good agreement with our results. Sparse oxide particles are visible in the exfoliated zone, suggesting renewed corrosion occurred after spallation. From this comparison, it can be deduced that the spalled oxide experienced depletion of Si and Cr before detachment, highlighting their critical role in oxide film fracture and detachment. This cyclic process of oxidation-exfoliation-reoxidation likely repeats throughout the corrosion and straining process.

In summary, the dissolution of Si-rich oxides in the surface oxide film generates a high density of defects, which consequently weakens the film's fracture resistance under sustained tensile stress. Given that the

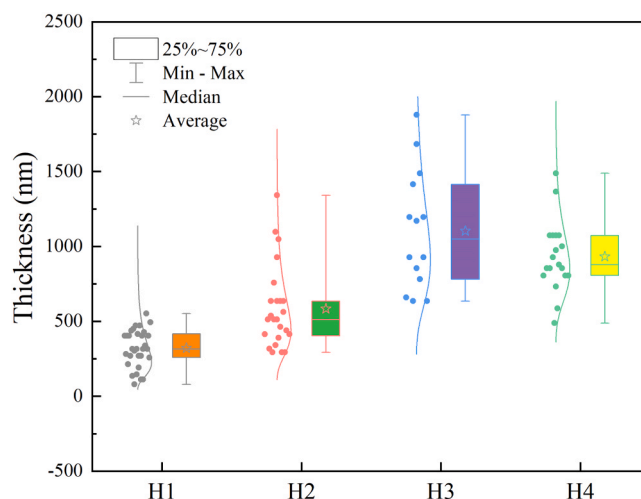


Fig. 9. Statistical distribution (box plot) of the inner oxide film thickness formed on four alloys. The data are fitted to a lognormal distribution.

oxidation mechanisms and resulting oxide structures are similar between the specimen surface and the GB, the GB oxides are expected to undergo an analogous process during crack initiation.

3.3.2. Cross-sectional view

Fig. 8 presents cross-sectional views of the oxide films and intergranular oxidation/cracks in the four alloys after SSRT. Due to the low crack density and large spacing between cracks, representative images of individual intergranular attacks are provided for each alloy. The cross-sectional morphology of oxide films is consistent with plan-view observations. As shown in Fig. 8a, alloy H1 exhibits sparse oxide particles, while alloy H4 shows a high density of oxide particles on the outermost surface. Cross-sectional analysis reveals that exfoliation occurs not directly at the metal-oxide interface, but within the inner oxide layer itself, resulting in oxide thinning. Alloys H1 and H4 exhibit only fine, vertically oriented cracks within the oxide. In contrast, alloys H2 and H3 developed horizontally propagating cracks that led to localized spallation. Additionally, numerous voids are observed within the upper oxide film on alloy H4, likely arising from continuous strain and dissolution.

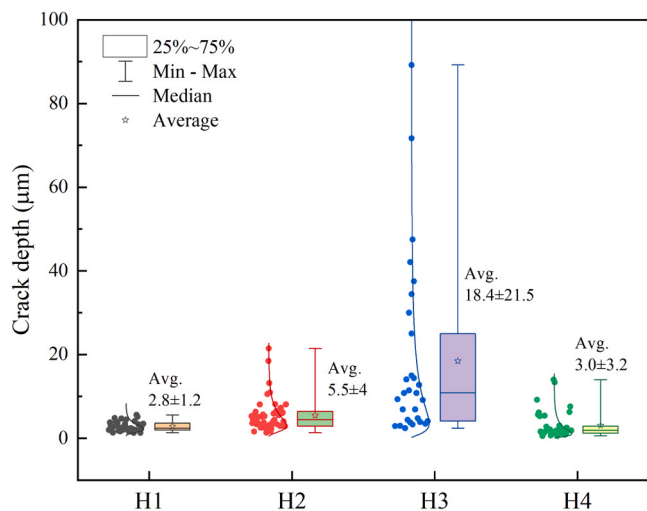


Fig. 10. Statistical distribution (box plot) of the cracking depth in alloys H1, H2, H3, and H4. The data are fitted to a lognormal distribution.

To evaluate the influence of Si and Cr on oxidation behavior, oxide thickness was statistically measured on polished side surfaces. Shen et al. [39] demonstrated that the composition and microstructure of surface oxides are similar to the SCC intergranular oxides; thus, the oxidation behavior of the surface can reflect intergranular oxidation. The measured thickness values and corresponding averages are summarized in Fig. 9. The data follow a lognormal distribution and exhibit considerable scatter in alloys H2 and H3, likely due to oxide spallation, yet a clear trend emerges: average oxide thickness increases significantly

with Si content. The thickness of the inner oxide layer generally reflects oxidation resistance and correlates well with SCC susceptibility in high-Si alloys; however, this trend does not hold for the low-Cr alloy (H4). Despite having a thicker average oxide layer than alloy H2, alloy H4 exhibits lower SCC susceptibility, indicating that uniform corrosion resistance alone does not determine cracking behavior. This discrepancy underscores the complexity of SCC mechanisms and suggests that differences in oxide evolution, such as stability under prolonged stress and susceptibility to fracture, play a critical role in determining overall SCC performance.

Fig. 8b shows representative intergranular oxidation/cracks for each alloy. Some GBs oxidation depths of up to several microns with a sharp wedge shape, which is consistent with the characterization of neutron-irradiated stainless steels that experience more severe GB oxidation than the unirradiated counterparts, with GB oxidation depth increasing with dose [36]. Part of the intergranular oxide cracked at a position close to the surface. Based on the characteristics of intergranular oxidation and cracks, it can be inferred that the cracks originate from preferentially oxidized GBs. In other words, GB oxidation is a precursor to intergranular cracking. This result is also consistent well with our previous study on neutron-irradiated 304 and 316 stainless steels [12, 13,40]. Our model alloys replicate the severe GB oxidation behavior of neutron-irradiated 304 and 316 stainless steels. Thus, the depth of intergranular oxidation/crack can reflect the IASCC susceptibility of alloys and was used to evaluate the cracking susceptibility of the model alloys.

Fig. 10 exhibits the depth distribution of intergranular cracks for each alloy, and the average crack depths are calculated and presented in the box plots. Due to the limited number of observed cracks, the distribution of intergranular crack depths was also fitted with a lognormal distribution. For the high-Si alloys, crack depth increases with Si

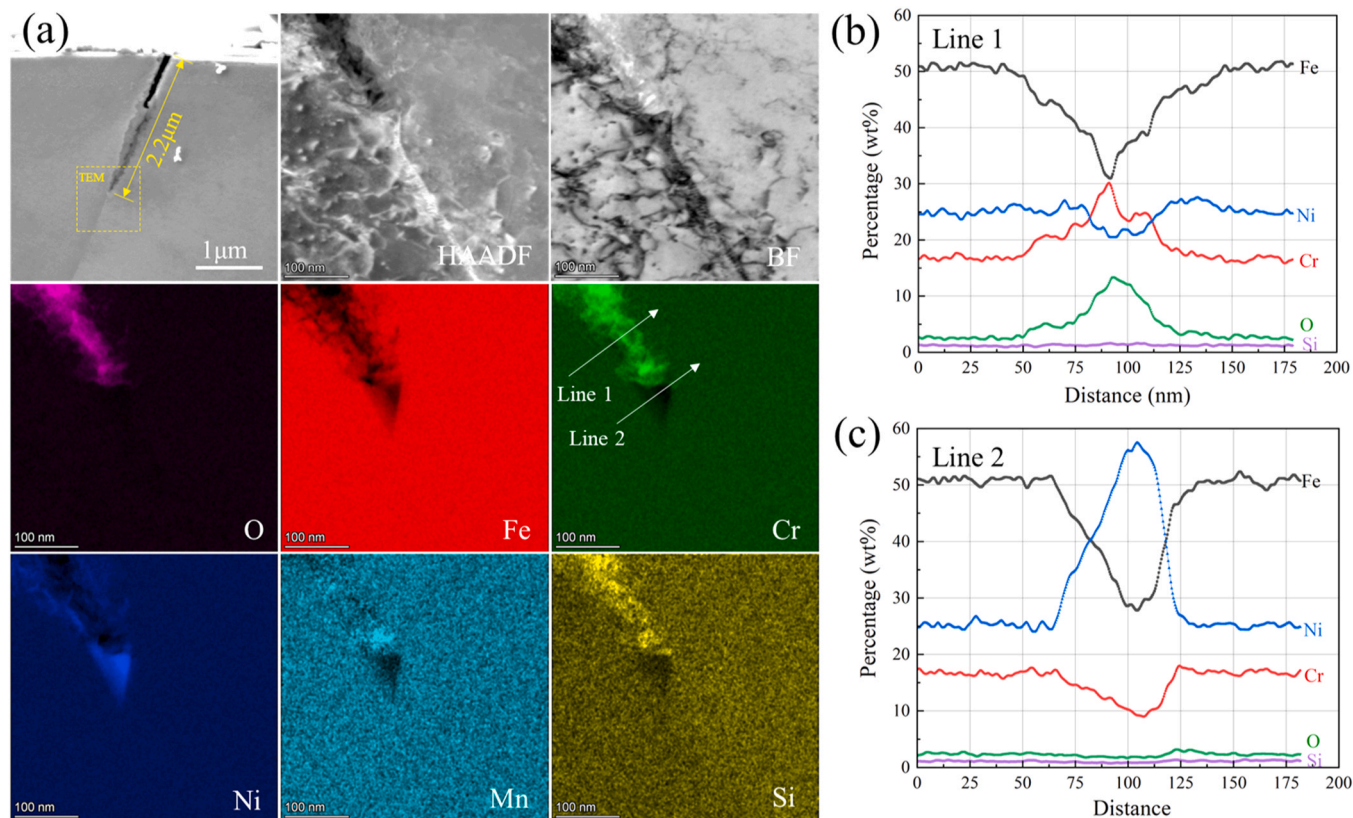


Fig. 11. STEM analysis of an intergranular oxide tip in alloy H1 after SSRT: (a) STEM-HAADF image and corresponding EDS elemental maps illustrating the oxide structure and composition; (b, c) EDS line scan profiles across the GB oxide (Scan Line 1) and ahead of the oxide tip (Scan Line 2), showing elemental variations near the oxidation front.

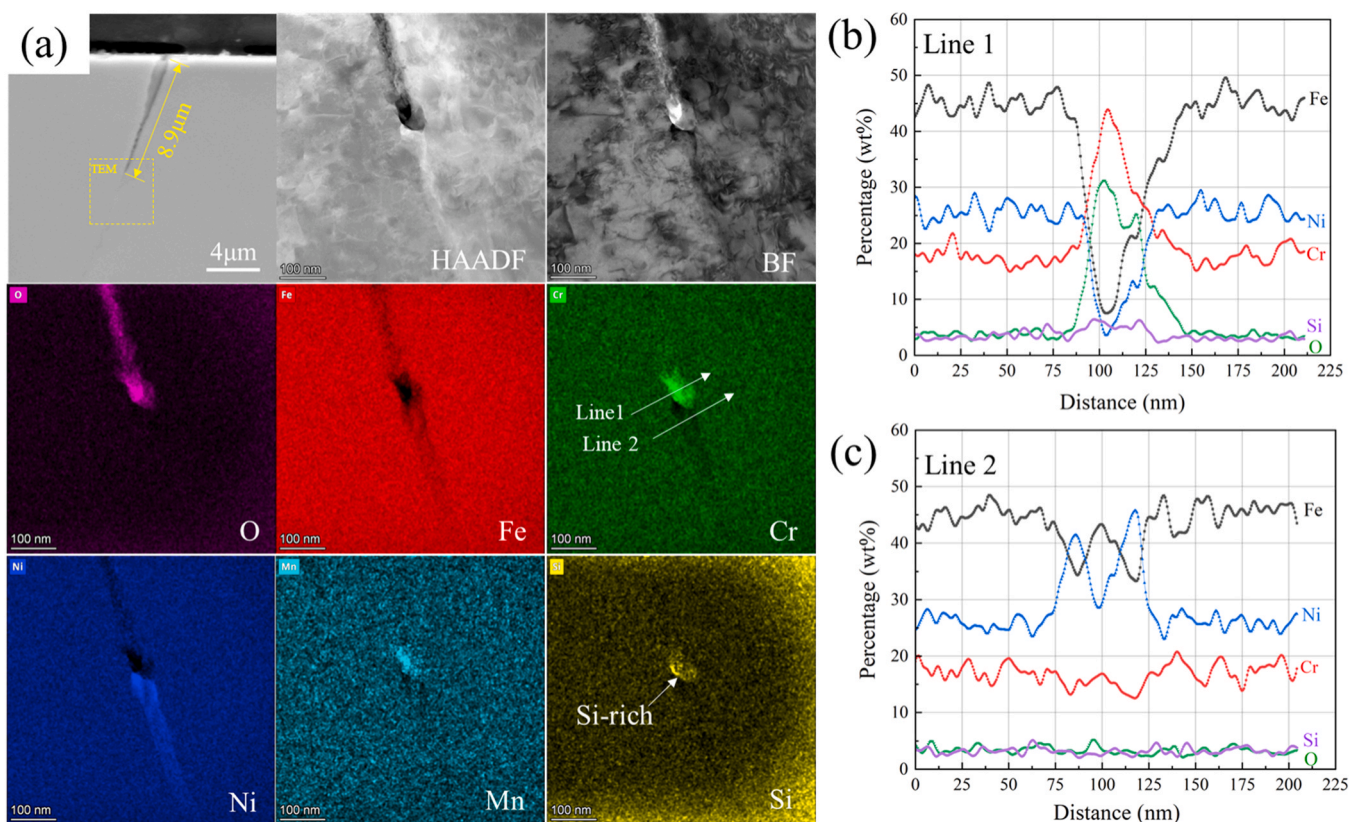


Fig. 12. STEM analysis of an intergranular oxide tip in alloy H2 after SSRT: (a) STEM-HAADF image and corresponding EDS elemental maps illustrating the oxide structure and composition; (b, c) EDS line scan profiles across the GB oxide (Scan Line 1) and ahead of the oxide tip (Scan Line 2), showing elemental variations near the oxidation front.

concentration, while the low-Cr alloy (H4) has a slightly larger depth than alloy H1. This result aligns well with the trends in crack length density, maximum crack length/width, and oxidation thickness. Although the average crack depths of alloys H1 and H4 are comparable, the maximum depth for alloy H4 is about 13 μm - far greater than the 5.5 μm maximum depth for alloy H1. This indicates that alloy H4 has higher SCC susceptibility than alloy H1, further supporting the prior conclusion. Among the Si-containing alloys, the maximum crack depth for alloy H3 reaches approximately 85 μm, much higher than that of alloys H1 and H2.

3.4. STEM characterization of crack tips

Intergranular oxidation and crack tips in alloys H1 to H4 were examined using STEM, with results presented in Figs. 11–14. Each figure includes an overview of the FIB-prepared sampled region, followed by HAADF, BF, and EDS mapping images of the tip region (Figs. 11–14a). A high density of dislocations, introduced by warm working or slow strain during testing, is evident surrounding the oxidation front in all alloys. Quantitative EDS line scans across the GB oxide and ahead of the tip are provided in Figs. 11–14(b, c).

For the high-Si alloys (H1, H2, H3), the oxidized GB tip consists of Cr- and Si-enriched oxide. The Si enrichment levels measured at the tip are 1.6 wt%, 6.4 wt%, and 17.5 wt% for alloys H1, H2, and H3, respectively, corresponding to 2.0, 2.5, and 4.4 times the matrix concentration. This indicates that Si enrichment at the tip escalates with increasing bulk Si content. Nickel enrichment is also observed ahead of the oxidized tip in these alloys. Additionally, GB migration, attributed to significant elemental diffusion [41,42], is notably present in H3. The length of the Ni-enriched zones is 58 nm, 239 nm, and 232 nm for alloys H1, H2, and H3, respectively. Although the Ni-enriched region is longer in alloy H2

than in H3, the higher enrichment ratio and observed DIMG in H3 suggest more extensive elemental outward diffusion. It should be noted that the anomalously high Si signal at one location in the EDS map of the H2 specimen is an artifact. A tungsten protective layer was deposited to preserve the GB oxide during FIB preparation of the TEM specimen. Therefore, the artifact is likely due to a peak overlap, specifically from the W-Kα line (1.775 keV), which has an energy very close to that of the Si-Kα line (1.740 keV).

The low-Cr alloy H4 exhibits distinct GB oxidation behavior compared to the high-Si alloys. No Si-rich oxide is formed along the GB. Nanoscale Cr-rich oxides are present in alloy H4, likely originating from pre-existing carbides. The Ni-enriched zone ahead of the oxide is narrow but exceeds 1.1 μm in length, with no accompanying GB migration. The apparent Si-rich region at the upper left corner of the Si map is also a mis-indexing of tungsten.

The carbides in this study are small and sparsely distributed. Although GB carbides have been reported to mitigate oxidation and slow SCC in hydrogenated water [43,44], their beneficial effect in this alloy is likely less pronounced than in alloys with intentionally introduced high-density carbides via thermal processing.

4. Discussion

Post-test analysis of surface oxide characteristics and crack morphology, combined with statistical analysis of crack density and depth, revealed distinct crack initiation behaviors among the four model alloys. As shown in Fig. 15, both crack length density and average intergranular crack depth increased across the alloy series. Crack length density is commonly associated with resistance to crack initiation, while crack depth reflects the tendency for crack propagation. Thus, the SCC susceptibility of the alloys, in descending order, is determined to be:

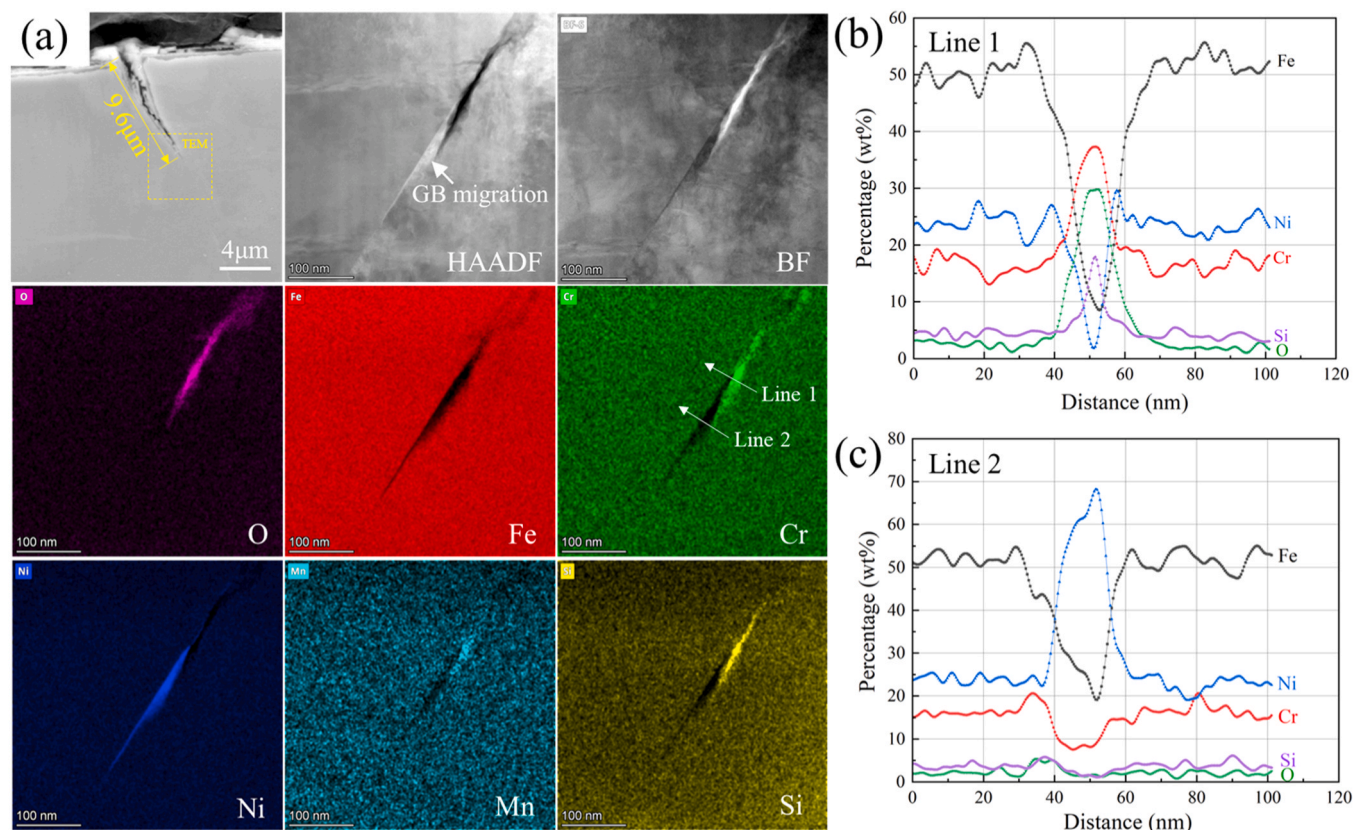


Fig. 13. STEM analysis of an intergranular oxide tip in alloy H3 after SSRT: (a) STEM-HAADF image and corresponding EDS elemental maps illustrating the oxide structure and composition; (b, c) EDS line scan profiles across the GB oxide (Scan Line 1) and ahead of the oxide tip (Scan Line 2), showing elemental variations near the oxidation front.

H3 > H2 > H4 > H1. For the high-Si alloys, SCC susceptibility increased with increasing Si concentration in the matrix. The low-Cr alloy exhibited comparatively low susceptibility, only marginally higher than that of alloy H1 (0.85 wt% Si). These results imply that in hydrogenated water the effect of Cr depletion at GBs on IASCC is secondary to the influence of Si enrichment, particularly when Si concentrations exceed 2.63 wt%.

The higher SCC susceptibility of the high-Si alloys compared to the low-Cr alloy can be attributed to differences in oxidation kinetics and oxide film evolution. The thickness of the surface oxide film was observed to increase with Si content, indicating an accelerated corrosion rate. This acceleration is likely driven by the oxidation of Si and subsequent degradation of the resulting Si-containing oxide film in high-temperature water. Silicon has a high affinity for oxygen and forms SiO_2 [45,46]. Higher Si concentrations lead to increased SiO_2 formation within the oxide layer. Although SiO_2 is protective in high-temperature air, it becomes unstable in aqueous environments and tends to dissolve preferentially [30,38,47–50], introducing defects into the oxide film. Higher SiO_2 content thus results in a more porous and less protective oxide structure, which facilitates continued corrosion and leads to thicker films. This porous oxide provides pathways for oxygen diffusion along GBs, promoting intergranular oxidation. Furthermore, the defective oxide is more prone to fracture under applied stress, which can trigger crack initiation. Consistent with this mechanism, oxide film exfoliation and Si depletion in the remaining oxide layers were observed in the high-Si alloys. STEM characterization of intergranular oxides and crack tips further confirmed that both GB diffusion rates and overall corrosion rates increase with Si content. In summary, the rapid degradation of the oxide film in high-Si alloys, driven by SiO_2 dissolution, accelerates corrosion and reduces the stress required for oxide fracture, thereby explaining the positive correlation between Si content and SCC

susceptibility.

In contrast, the low-Cr alloy (H4) exhibited low SCC susceptibility, likely due to the slower degradation of its oxide film. Compositional analysis suggests that its inner oxide layer is similar to those of the high-Si alloys but with lower Cr and Si fractions [38]. Chromium is known to reduce corrosion rates by forming a protective Cr-rich oxide that hinders ion diffusion [51]. The composition and thickness of the inner oxide layer depend strongly on the matrix Cr content, with higher Cr levels leading to higher Cr fractions in the oxide and thinner films. The measured oxide thickness on alloy H4 ($\sim 0.93 \mu\text{m}$) is consistent with its low Cr content and aligns with findings from Terachi et al. [51]. Despite developing the second-thickest oxide film after H3, alloy H4's inner oxide layer, which had a low SiO_2 content, retained its integrity and resisted transformation into a porous structure. Thus, despite its thickness, the oxide on alloy H4 retained resistance to ion diffusion and mechanical fracture, unlike the brittle oxide films on high-Si alloys.

However, based on the above analysis, one might expect alloy H4 to have the lowest SCC susceptibility, given its lower Si content compared to H1 (0.85 wt% Si). This discrepancy is likely explained by the much higher Cr content in H1 (19.84 wt%) and, consequently, the formation of higher Cr containing inner oxide, which mitigates the detrimental effects of SiO_2 dissolution.

Based on these findings, we conclude that Si enrichment at GBs, rather than Cr depletion, is a more important factor contributing to high IASCC susceptibility in hydrogenated high-temperature water. Therefore, controlling Si concentration at low levels is essential for developing advanced IASCC-resistant alloys. It should be noted, however, that these conclusions are environment-specific. Since this study was conducted in hydrogenated water, further validation in oxygenated conditions is necessary, as different oxidation and dissolution behaviors may alter the relative contributions of Si and Cr. Many researchers have reported that

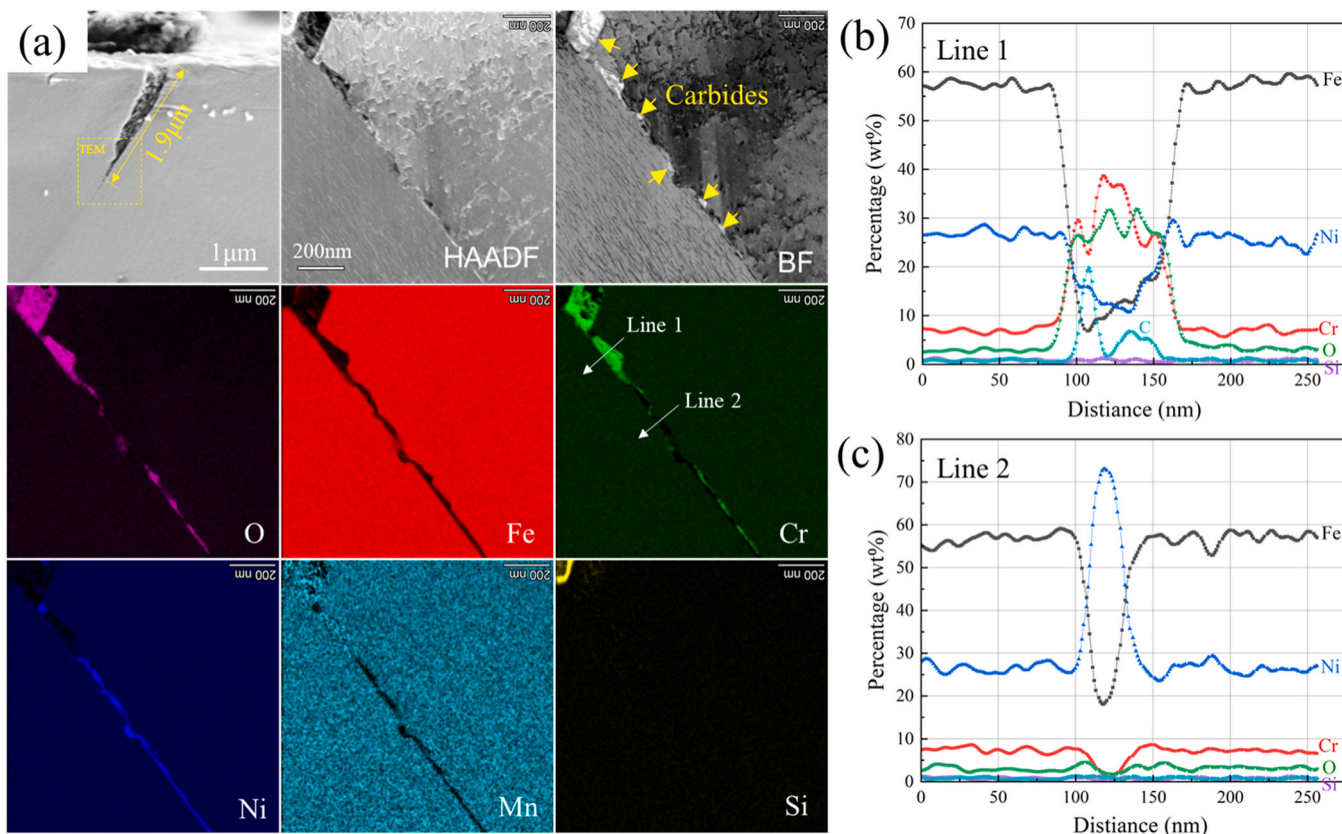


Fig. 14. STEM analysis of an intergranular oxide tip in alloy H4 after SSRT: (a) STEM-HAADF image and corresponding EDS elemental maps illustrating the oxide structure and composition; (b, c) EDS line scan profiles across the GB oxide (Scan Line 1) and ahead of the oxide tip (Scan Line 2), showing elemental variations near the oxidation front.

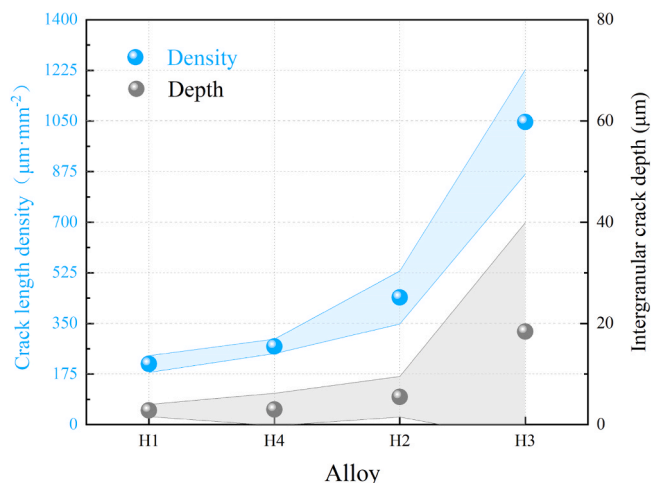


Fig. 15. Crack length density and intergranular cracking depth for each alloy after the SSRT.

the Cr-rich oxide tends to be dissolved in oxygenated water [52,53]. Previous studies indicate that both Cr depletion and SiO_2 dissolution play more pronounced roles in oxygenated environments [21,46,54], which may lead to different trends.

An additional observation in the low-Cr alloy was the presence of fine carbides on GBs. Although their small size and sparse distribution likely limited their impact on oxidation resistance compared to intentionally introduced thermal carbides, future studies using carbide-free model alloys are still recommended to eliminate any potential

confounding effects of carbides.

5. Conclusion

To clarify the respective roles of GB Si enrichment and Cr depletion in IASCC, four model alloys with systematically varied Si and Cr concentrations were subjected to slow strain rate testing in simulated hydrogenated high-temperature water, with testing stopped once 7 % plastic strain was achieved. Distinct differences in crack initiation behavior were observed, and these differences were correlated with insights from comprehensive microstructural characterization. The main findings are as follows:

- 1) SCC susceptibility in hydrogenated water increases significantly with increasing Si content, while showing low dependency on Cr content. This indicates that Si segregation at GBs, rather than Cr depletion, is a more important factor controlling crack initiation in such environments. Cr depletion alone is insufficient to induce substantial IASCC.
- 2) The elevated SCC susceptibility of high-Si alloys is attributed to the dissolution of SiO_2 in the inner oxide layer, which leads to the formation of a porous and defective oxide film. This degraded oxide film structure accelerates intergranular oxidation and compromises the mechanical integrity of the film, thereby facilitating crack initiation under applied stress.
- 3) Based on these results, it is recommended that Si content be maintained at a much lower level when developing advanced IASCC-resistant structural materials for nuclear reactor core components.

CRediT authorship contribution statement

DU Donghai: Writing – original draft, Visualization, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Hefei Huang:** Writing – review & editing, Validation, Investigation, Formal analysis. **Tenghong Lin:** Writing – review & editing, Investigation, Data curation. **Lupeng Liu:** Visualization, Investigation, Data curation. **Shuai Chen:** Investigation, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

The authors do not have permission to share data.

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